Android Project: Trip Organizer Application

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Objective

The objective of this project is to design and develop an Android application called “Trip Organizer”, which helps users plan a tourist trip by managing a list of places to visit.

The application must use SQLite for data persistence and implement various Android features such as navigation, intents, localization, user preferences and others.

Application Description

The application allows the user to create and manage a single trip.

A trip is defined by:
- A city
- A start date
- An end date

Each trip contains a list of places to visit.

A place is characterized by:
1. A unique identifier
2. A title and a description
3. A date
4. A time (optional)
5. An address
6. A phone number (optional)
7. A representative photo (optional)
8. A visited status (boolean: visited / not visited)

Data Model

- One Trip contains multiple Places
- The application manages only one active trip at a time
- When a trip is deleted, all associated places must also be deleted

Functional Requirements

Trip Management:
- Create a new trip (city, start date, end date)
- Edit a trip
- Delete the current trip and all its associated places

Place Management :
- Add a new place
- Display the list of places
- Update an existing place
- Delete a place

Application Features

For each place displayed in the list, the user must be able to:
- View detailed information
- Modify the place
- Mark the place as visited / not visited
- Open the address in Google Maps
- Open the phone dialer with the phone number (if provided)
- Share the place information using an Android share intent

Photo Handling

Each place may include a photo selected from the device.
- The user can choose an image from the device storage
- The selected image should be displayed in the application
- The image reference must be saved and reloaded correctly

Data Persistence

All data must be stored locally using SQLite.
The application must correctly handle:
- Insert
- Select
- Update
- Delete

User Interface Requirements

The application should include at least:
- One screen to create or edit the trip

- One screen to display the list of places
- One screen to add, view or edit a place

Some Technical Requirements

- Use DatePicker and TimePicker for date and time selection
- Use a ListView (or RecyclerView) to display places
- The list should display relevant information (e.g., title, date, visited status)
- Optional fields must be handled properly (no crashes if empty)

User Preferences

The application must include a settings screen allowing the user to manage preferences.

- Background music: ON / OFF
- Notification: 1 week before/ 2 days before

Preferences must be persisted locally and applied when the application starts

Background Music (Service)

The application must include a background music feature.

- A music track should be played when the application is running
- The music must respect the user preference (ON / OFF)
- The music handling should be implemented using an Android Service

Expected behavior:
- Music starts when the application is opened (if enabled)
- Music stops when the application is closed or disabled in settings

Notifications

The application must raise a notification according to user preferences: 1 week or 2 days before the trip.

Validation and Business Rules

Trip Validation:
- The city must not be empty
- The start date and end date must be provided
- The start date must be earlier than or equal to the end date

Place Validation:
- A place must include at least a title, a description, a date, and an address
- Optional fields (time, phone number, photo) may be left empty, but must not cause any application crash
- If a phone number is provided, it must be valid (basic validation is sufficient)

Consistency Rules:
- The date of a place must be within the trip period (between the start and end date)

Trip Modification Rules:
- The trip can be modified only if no places have been added yet
- Once at least one place has been added, the trip information can no longer be modified

Localization

The application must support two languages:
- French
- English

The displayed language must depend on the device's locale

Bonus or Optional Extension (Advanced – Multiple Trips)

The application currently manages a single trip at a time.

As an advanced feature, you may extend the application to support multiple trips.

In this case, the application should:
- Allow the user to create and manage multiple trips
- Display a list of trips
- Allow selecting a trip to view its associated places
- Ensure that each place is linked to a specific trip

This feature requires adapting both the data model and the user interface.
It will be considered for higher evaluation.

Submission Requirements

Your submission must include a single ZIP file containing:

1. A small report (PDF) including:
   - Description of your approach
   - Problems encountered

- Implemented and non-implemented features
- Screenshots of the application

2. The complete source code (Android Studio project)

3. A video demonstration of the application

Notes

- The application should be functional and user-friendly
- Code should be clear and well-structured
- All required features should be implemented